



DAYANANDA SAGAR UNIVERSITY

Devarakaggalahalli, Harohalli, Kanakapura Road, Bengaluru South Dt. – 562 112

**Bachelor of Technology
in**

COMPUTER SCIENCE AND ENGINEERING

Major Project Phase-II Report

**CALM WITH CARNATIC: AN AI-DRIVEN FRAMEWORK FOR
MUSIC THERAPY AND STRESS MONITORING**

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CERTIFICATE

This is to certify that the Phase-II project work titled “**CALM WITH CARNATIC: AN AI-DRIVEN FRAMEWORK FOR MUSIC THERAPY AND STRESS MONITORING**” is carried out by **Sampreet Kishan PK (ENG22CS0436), Shiva Karthick S (ENG22CS0454), Sriharsha Chalamacharla (ENG22CS0468), Shwetangi Thakur (ENG22CS0462)** bonafide students of Bachelor of Technology in Computer Science and Engineering at the School of Engineering, Dayananda Sagar University, Bangalore in partial fulfillment for the award of degree in Bachelor of Technology in Computer Science and Engineering, during the year **2025-2026**.

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LIST OF ABBREVIATIONS

EEG	Electroencephalogram
MFCC	Mel-frequency Cepstral Coefficients
ZCR	Zero-Crossing Rate
CNN	Convolutional Neural Network
RAVDESS	Ryerson Audio-Visual Database of Emotional Speech and Song

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ABSTRACT

Stress and emotional dysregulation are prevalent issues lacking accessible, real-time therapeutic interventions. This project introduces "Raga Chikitsa," a dual-pipeline web application that prescribes personalized Carnatic music therapy based on real-time neural and vocal biomarkers. The first pipeline captures live EEG data via Bluetooth hardware, calculating Alpha/Beta brainwave ratios to assess cognitive states. The second pipeline extracts Mel-frequency cepstral coefficients (MFCCs) from vocal recordings, utilizing a Convolutional Neural Network (CNN) to classify emotional states with high accuracy. Based on these objective physiological inputs, the system dynamically maps and streams specific therapeutic Ragas (such as Ananda Bhairavi or Bilahari) targeted to the user's current psychological baseline. Built on a React frontend and FastAPI backend, the architecture also features secure Google OAuth integration for automated clinical session logging to Google Drive. Ultimately, this system bridges ancient sound healing therapies with modern machine learning and neuro-informatics, providing an accessible, data-driven tool for mental wellness and personalized music therapy.

CHAPTER 1

INTRODUCTION

CHAPTER 1 INTRODUCTION

1.1 INTRODUCTION:

Stress is a physiological and anatomical response that, when left untreated, can lead to severe health complications such as heart disease and depression. While traditional treatment methods exist, they are often reactive and inaccessible during acute moments of distress, highlighting a critical need for home-based therapy systems capable of offering immediate relief.

Recent advancements in *electroencephalography* (EEG) offer reliable methods to monitor mental health by analyzing the power spectral density (PSD) of brainwaves. Specifically, the ratio between the **Alpha (8-13 Hz)** and **Beta (13-30 Hz)** bands provides a highly accurate indicator of cognitive load and stress levels. Concurrently, traditional Indian Carnatic Music Therapy, known as "**Raga Chikitsa**" operates on the principle that specific melodic scales can modulate physiological parameters, such as pulse rates and anxiety levels.

This project bridges the gap between biomedical engineering and traditional music therapy by developing an accessible, full-stack web application titled "**Calm with Carnatic**". The system monitors mental health through two primary modalities:

1.1.1. SHORT-TERM ANXIETY RELIEF:

Utilizing a **BrainLink** Lite EEG headset, the system records the Alpha/Beta ratio in real-time. If the ratio drops below a healthy threshold, indicating stress, the system automatically intervenes by playing a calming Carnatic Raaga, such as Ananda Bhairavi.

1.1.2. LONG-TERM EMOTIONAL SUPPORT:

For users without specialized hardware, the system employs an audio-based emotion recognition module. By extracting audio features and processing them through a **1D Convolutional Neural Network (1D-CNN)**, the system classifies the user's vocal emotion with 82% accuracy and prescribes mood-elevating Raagas like Bilahari.

1.2 OBJECTIVES:

1.2.1 DEVELOP A REAL-TIME BIOFEEDBACK SYSTEM:

To capture and analyze live EEG signals using the BrainLink Lite v2.0 headset, specifically calculating the Alpha/Beta brainwave ratio to accurately detect acute stress.

1.2.2 IMPLEMENT VOCAL EMOTION RECOGNITION MODEL:

To extract Mel-Frequency Cepstral Coefficients (MFCCs) and other spectral features from user voice inputs, classifying their emotional state using a 1D Convolutional Neural Network (1D-CNN) trained on the RAVDESS and TESS datasets.

1.2.3 AUTOMATE MUSIC THERAPY (RAGA CHIKITSA):

To dynamically map the detected physiological and emotional states to specific, scientifically validated Carnatic Raagas (e.g., Ananda Bhairavi for stress, Vakulabharanam for anger) to provide immediate therapeutic relief.

1.2.4 DEPLOY A FULL-STACK ARCHITECTURE:

To integrate the standalone Python EEG and Audio analysis modules into a unified, accessible web application using a React frontend and FastAPI backend, enabling secure clinical session logging to Google Drive.

CHAPTER 2

PROBLEM DEFINITION

CHAPTER 2 PROBLEM DEFINITION

2.1 THE GLOBAL BURDEN OF STRESS:

Stress is a physiological response that, when unmanaged, leads to severe chronic conditions like cardiovascular disease and depression. High-pressure environments subject individuals to heavy distress daily, yet immediate and effective relief remains largely inaccessible when people need it most.

2.2 LIMITATIONS OF TRADITIONAL APPROACHES:

Traditional psychiatric treatments fundamentally suffer from being reactive rather than proactive. They require scheduling and forethought, making them useless during acute, real-time moments of distress. Furthermore, they fail to provide a continuous, non-invasive method for tracking an individual's emotional degradation over time without expensive medical supervision.

2.3 GAP IN TECHNO-THERAPEUTIC INTERVENTIONS:

There is a massive gap between complex biomedical engineering and practical, accessible treatment like traditional Indian Carnatic Music Therapy ("Raga Chikitsa"). The core problem this project solves is the lack of a unified, automated framework that can simultaneously intercept acute anxiety using EEG hardware and provide long-term emotional regulation using accessible vocal recognition software.

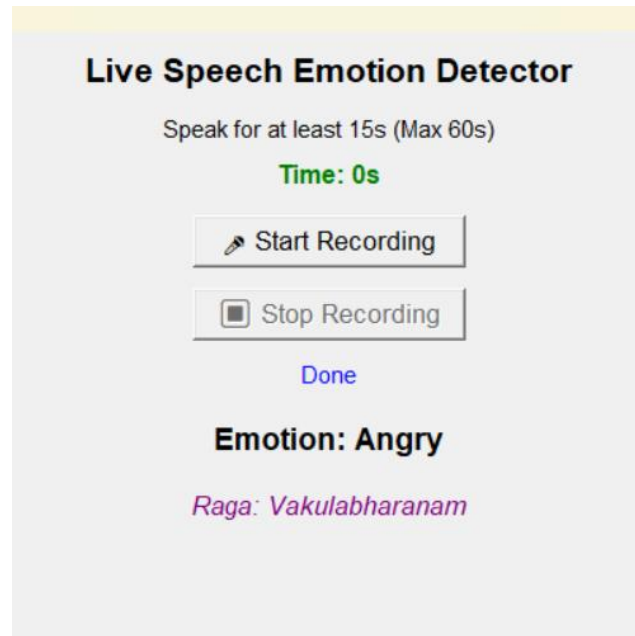


FIGURE 2.1 GUI playing Vakulabharanam to calm a person in Angry state

CHAPTER 3

LITERATURE SURVEY

CHAPTER 3 LITERATURE SURVEY

Author's name / Paper's Title	Conference name	Technology/Design	Results by Author	What we Infer
"Emotion Recognition on Speech Signals Using Machine Learning", Mohan Ghai, Shamit Lal, Shibam Duggal and Shrey Manik	IEEE 2017	SVM, MFCC Extraction, Random Forest	The paper shows the adding energy to MFCC features and using overlapping segments greatly improves speech emotion recognition.	This paper helps us understand that MFCC is one of the main audio features by which emotion can be deduced and how to extract the said feature.
"Sound based human emotion recognition using MFCC and Multiple SVM" Anagar Sonawane, M.U Inamdar and Kishor B. Bhangale	IEEE 2017	Automatic Speech Recognition, SVM, Speech Emotion Recognition	Motion Sound features are extracted using MFCC and classification is done using linear and nonlinear multiple SVM	This helps us understand that SVM is a good emotion classifier and can be done using linear and nonlinear data.
"Electroencephalogram stress analysis on alpha/beta ratio and theta/beta ratio" Tee Yi Wen, Siti Armiza Mohd Aris	Indonesian Journal of Electrial Engineering and Computer Science 2020	EEG, Power Ratio, Power Spectral Density, Stress, Virtual Reality	The study shows that Alpha/Beta EEG power ratios reliably distinguish relaxed and stressed states, supporting their use for future stress level classification on systems.	Alpha/Beta ratio can be used to distinguish the stress levels to classify amongst them.
"Emotion impact of four specific ragas of carnatic music" Dr. Meenakshi Ravi, Miss. Sharnya Govindaraj	International Journal of Alternative and Complementary Medicine 2021	Indian Music Therapy, Carnatic Music, Raga Therapy	Khanada and Vakulabharanam ... Can be beneficial	This helps us to classify the ragas for different levels of stress and gives details on which ragas help different individuals and mental states.

“Indian Classical Music, Raaga as music Therapy: Scope and Oppotunities” Varun Malhotra, Samidha Vedabala, Sagar Khadanaga, Anvesh Jallapally, Murlimanju BV	Manipal Journal of Medical Sciences 2022	Music Therapy, Emotion, Raga and Emotions, Indian Music	Indian classical music is long believed to support healing and enhance well-being through elements like notes, pitch, tempo and tone variation	Indian classical music can be used to support healing and enhance the patient’s physical and mental state.
“Effect of Carnatic raga- Bilahari based music therapy on anxiety, sleep and somatic symptoms among caregivers of cancer patients” Ramachandran Krishan, Eslavath Rajkumar, John Romate, Joshua George Allen, Daniel Monica	Heliyon 2022	Caregivers, Music Intervention, Anxiety, Sleep, Somatic Symptomss	Bilahari Raga significantly improves cancer caregivers' mental health by reducing anxiety, somatic symptoms, and sleep issues. It suggests that this low-cost music intervention could also benefit other vulnerable groups facing stress or emotional strain.	This study provides information regarding the improvements of the mental well-being of cancer patients after listening to Bilahari Raga.

Table 3.1 Literature Survery

CHAPTER 4

PROJECT DESCRIPTION

CHAPTER 4 PROJECT DESCRIPTION

4.1 PROPOSED DESIGN:

The proposed "Raga Chikitsa" framework is engineered as a comprehensive, dual-modal architecture designed to deliver automated, non-invasive mental health interventions. The system processes physiological and acoustic biomarkers through two distinct but complementary pipelines to offer both acute stress relief and long-term emotional regulation.

4.1.1 EEG PIPELINE (ACUTE STRESS MANAGEMENT):

The first modality utilizes the BrainLink Lite v2.0 headset, a single-channel Brain-Computer Interface (BCI) equipped with a dry electrode positioned on the pre-frontal cortex. Raw electroencephalogram (EEG) signals are transmitted via a Bluetooth 4.0 interface at a sampling rate of 512 Hz. A custom parsing algorithm processes this data stream, applying a 50 Hz filter to eliminate power line interference, and utilizes Welch's method to compute the Power Spectral Density (PSD). The core mechanism continuously calculates the ratio of Alpha (8-13 Hz) to Beta (13-30 Hz) brainwave power. If this ratio drops below the 1.0 threshold (indicating acute anxiety and cognitive overload), the system automatically triggers the playback of Ananda Bhairavi to aggressively reduce physiological stress markers. Ratios between 1.0 and 1.3 trigger Bilahari to maintain calmness, while ratios above 1.3 trigger Shankarabharanam to promote a majestic, balanced state.

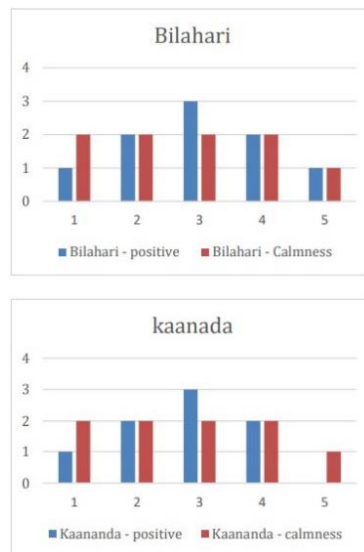


Figure 4.1(a) Bilahari and Kaanada results on positivity/calmness for people

4.1.2 VOCAL LONG-TERM TRACKING:

The second modality serves users without specialized EEG hardware by analyzing vocal biomarkers. The system records short audio segments and utilizes advanced digital signal processing to extract a rich, 218-dimensional acoustic feature vector. This array includes Mel-Frequency Cepstral Coefficients (MFCCs), Chroma STFT, Zero Crossing Rate (ZCR), Tonnetz, and RMS Intensity. These features are fed into a custom 1D Convolutional Neural Network (1D-CNN) architecture. This deep learning model categorizes the user's emotional state into one of five discrete classes: Neutral, Happy, Sad, Angry, or Fearful. Based on this classification, the system dynamically prescribes mood-altering Carnatic Raagas, such as *Thodi* for sadness or *Vakulabharanam* for anger, acting as an automated mood journal.

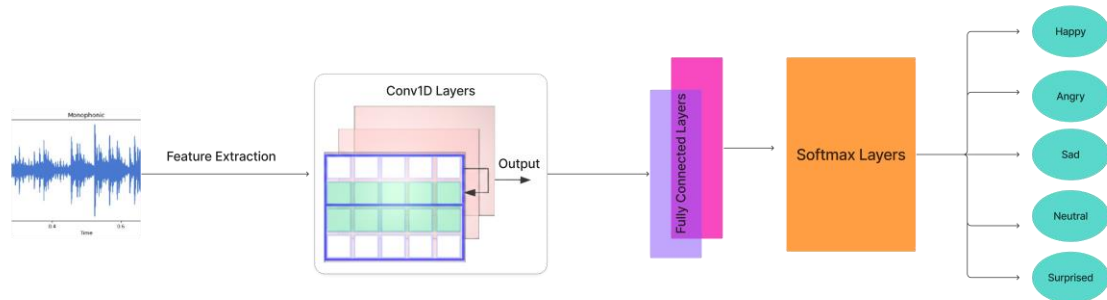


Figure 4.1(b) 1D CNN classifying emotions into five categories

4.1.3 FULL-STACK SYSTEM INTEGRATION:

Both standalone pipelines are integrated into a modern Full-Stack Web Application architecture. A responsive React-based Graphical User Interface (GUI) communicates with a FastAPI Python backend via WebSockets and HTTP endpoints. This allows for the real-time visualization of brainwaves and automated, secure logging of therapy sessions to the cloud (Google Drive) for long-term clinical review.

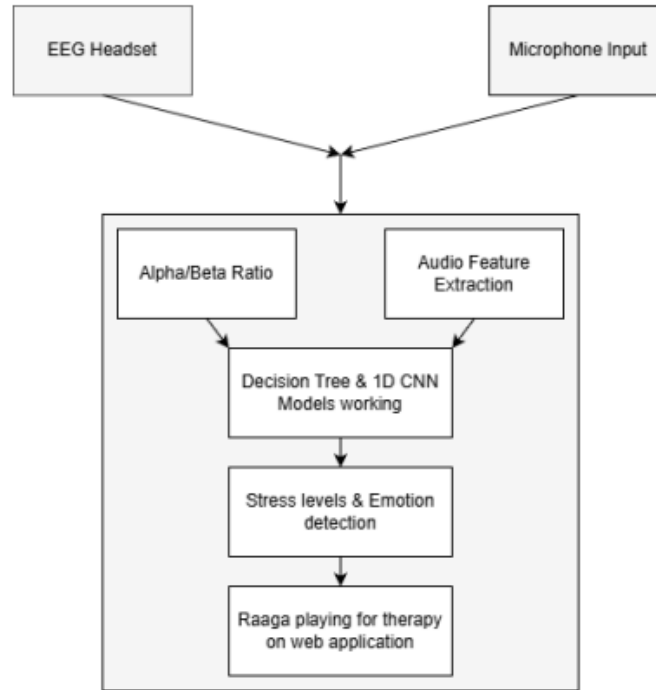


Figure 4.1(c) High Level Design of the Full Stack Web Application

4.2 ASSUMPTIONS AND DEPENDENCIES:

The successful deployment, execution, and accuracy of this framework rely on several fundamental hardware and software dependencies.

4.2.1 HARDWARE DEPENDENCIES:

The acute stress monitoring module fundamentally assumes the availability and correct physical placement of the BrainLink Lite v2.0 EEG headset on the user's forehead. Furthermore, the system depends on a highly stable Bluetooth 4.0 connection to ensure the uninterrupted, real-time transmission of raw brainwave packets without critical data loss or latency. The vocal analysis pipeline assumes the presence of a standard, high-fidelity microphone capable of capturing clear acoustic signals free from excessive environmental noise, which is an absolute necessity for accurate MFCC extraction.



Figure 4.2(a) BrainLink v2.0 is a version of a medical grade EEG system.

4.2.2 SOFTWARE DEPENDENCIES:

The backend mathematical processing and machine learning inference strictly require a robust Python 3.11 environment. This specific environment dependency is critical to circumvent legacy "Kernel shape" mismatches and serialization errors previously observed when loading pre-trained .h5 Keras models. The system also assumes the correct installation and configuration of essential external libraries, specifically **Librosa** for acoustic signal processing and **TensorFlow** for the execution of the 1D-CNN.

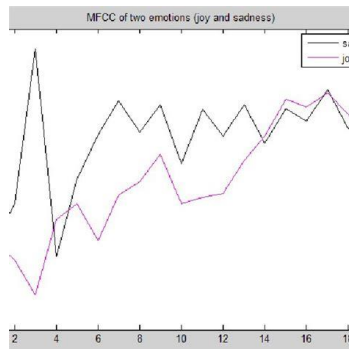


Figure 4.2(b) MFCC Extraction of joy and sadness by the 1D CNN Model

4.2.3 CONNECTIVITY AND AUTHENTICATION:

For the long-term clinical logging functionality, the system assumes an active internet connection to securely authenticate the user via Google OAuth 2.0. It depends on the Google Drive API to successfully export and push the finalized CSV session data to the cloud.

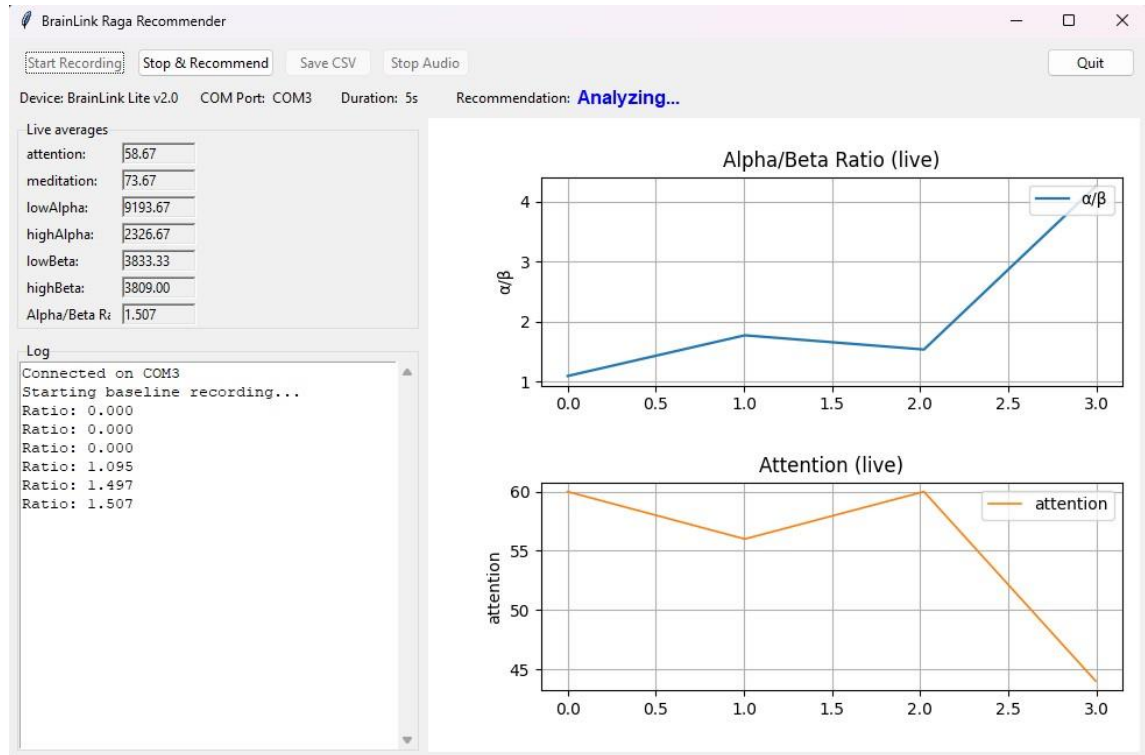


Figure 4.2(c) Raaga Bilahari playing to maintain Neutral State

CHAPTER 5

REQUIREMENTS

CHAPTER 5 REQUIREMENTS

The successful execution of the "Raga Chikitsa" framework relies on a strict set of functional capabilities and specific technological dependencies. These requirements are divided into functional expectations of the dual-pipeline system and the exact software and hardware stacks required for deployment.

5.1 FUNCTIONAL REQUIREMENTS:

The system must satisfy the following operational criteria to ensure real-time therapeutic intervention:

- **Real-Time EEG Data Acquisition:** The system must establish and maintain a secure Bluetooth 4.0 connection to the external EEG headset to acquire raw brainwave data at a continuous sampling rate of 512 Hz.
- **Neuro-Signal Processing:** The backend must parse the proprietary packet stream and apply a 50 Hz notch filter to eliminate environmental power line interference. It must subsequently apply a Hamming window-based band-pass filter (0.5-30 Hz) to isolate the Alpha and Beta frequency bands.
- **Acoustic Feature Extraction:** The vocal analysis pipeline must be capable of recording user audio and extracting a comprehensive 218-dimensional feature vector. This extraction must compute Mel-Frequency Cepstral Coefficients (MFCCs), Chroma STFT, Zero Crossing Rate (ZCR), Tonnetz, and RMS Intensity.
- **Algorithmic Classification & Intervention:** The system must compute the Welch Power Spectral Density (PSD) to dynamically calculate the Alpha/Beta ratio. Concurrently, it must pass the extracted acoustic features through a 1D Convolutional Neural Network (1D-CNN) to classify the user's emotion into one of five states: Neutral, Happy, Sad, Angry, or Fearful.
- **Automated Audio Mapping:** Upon detecting a physiological stress marker ($\rho < 1.0$) or a negative emotional state (e.g., Sadness, Anger), the system must autonomously trigger the playback of a corresponding therapeutic Carnatic Raaga (e.g., *Ananda Bhairavi*, *Thodi*, *Vakulabharanam*).
- **Session Logging:** The system must allow users to authenticate securely and export their session data (Alpha/Beta ratios, detected emotions, and prescribed Raagas) to a cloud storage environment for long-term clinical tracking.

5.2 SOFTWARE REQUIREMENTS:

To circumvent legacy system errors—such as "Kernel shape" mismatches and library version conflicts—the system strictly requires a modern, containerized software stack.

- **Core Backend Environment:** Python 3.11.
- **API & Networking:** FastAPI for high-performance HTTP endpoint routing and Uvicorn for ASGI server hosting. WebSockets must be utilized for the real-time, bidirectional streaming of live EEG data to the client interface.
- **Machine Learning & Audio Processing:** TensorFlow and Keras for hosting and executing the 1D-CNN inference model. Librosa, NumPy, and Soundfile for advanced digital audio signal processing and MFCC feature extraction.
- **Frontend User Interface:** A modern web application built using React.js and JavaScript, utilizing @react-oauth/google for secure user authentication and Google Drive API integration.

5.3 HARDWARE REQUIREMENTS:

The physical data acquisition layer requires the following specialized equipment:

- **EEG Biofeedback Device:** A BrainLink Lite v2.0 headset functioning as a single-channel Brain-Computer Interface (BCI). It must feature a dry electrode sensor capable of stable placement on the user's pre-frontal cortex and utilize a Bluetooth 4.0 dongle for transmission.
- **Acoustic Input:** A standard computer or mobile microphone capable of capturing clear, noise-free vocal samples.

CHAPTER 6

IMPLEMENTATION APPROACH

CHAPTER 6 IMPLEMENTATION APPROACH

6.1 MODULAR SYSTEM ARCHITECTURE:

The development strategy for the "Raga Chikitsa" framework prioritized a modular architecture, intentionally isolating the hardware-driven biofeedback loop from the software-driven audio classification system. This decoupling ensures that the computationally heavy 1D Convolutional Neural Network (1D-CNN) inference does not block the real-time, high-frequency WebSocket streaming of the EEG data. Both modules operate efficiently as standalone processes before converging into the centralized FastAPI web server.

6.2 EEG PROCESSING AND BIOFEEDBACK LOGIC:

The acute stress management pipeline relies on precise digital signal processing (DSP). Data acquisition initiates via a Bluetooth 4.0 COM port connection to the BrainLink Lite v2.0 headset, receiving data at a sampling rate of 512 Hz. A custom `BrainLinkParser` algorithm decodes the proprietary byte stream.

Once acquired, the raw signal undergoes immediate pre-processing. A 50 Hz notch filter is applied to eliminate AC power line noise, followed by a Hamming window-based band-pass filter (0.5–30 Hz) to isolate the target Alpha and Beta frequency bands. The core mathematical computation utilizes Welch's method to calculate the Power Spectral Density (PSD). The system continuously derives the Alpha/Beta power ratio (ρ).

The automated intervention logic operates on strict physiological thresholds:

- **$\rho < 1.0$ (Stress State):** The system triggers the playback of *Ananda Bhairavi*.
- **$1.0 \leq \rho \leq 1.3$ (Moderate/Neutral State):** The system triggers *Bilahari*.
- **$\rho > 1.3$ (Maintenance State):** The system triggers *Shankarabharanam*.

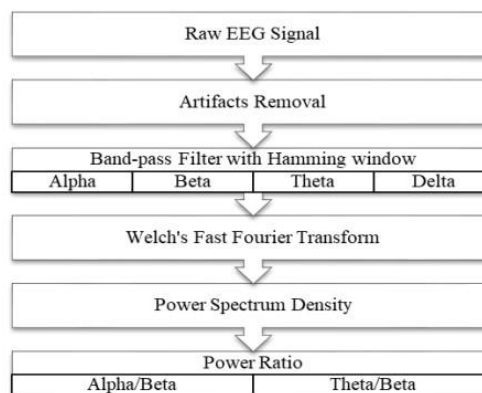


Figure 6.2 Raw EEG to Alpha/Beta Power Ratio Values

6.3 VOCAL EMOTION RECOGNITION (VER) PIPELINE:

The long-term emotion tracking module leverages advanced acoustic feature engineering. Utilizing the Librosa digital audio library, the system processes short vocal recordings to extract a comprehensive 218-dimensional feature vector. This multi-dimensional array includes 32 Mel-Frequency Cepstral Coefficients (MFCCs), Chroma STFT, Zero Crossing Rate (ZCR), Tonnetz, and RMS Intensity.

This vector is fed into a custom 1D Convolutional Neural Network (1D-CNN). The 1D-CNN architecture was explicitly selected because convolutional layers efficiently capture local spectral patterns and temporal dependencies in audio signals without suffering from the dimensionality curse inherent in older models like Support Vector Machines (SVM). The network terminates in a fully connected layer followed by a Softmax activation function, which outputs probabilities to classify the user's emotion into one of five distinct states: Neutral, Happy, Sad, Angry, or Fearful.

6.4 FULL-STACK AND CLOUD LOGGING:

To transition the framework from isolated algorithms to a deployable clinical tool, the modules are wrapped in a modern web architecture. The frontend GUI is constructed using React, featuring dedicated views for live brainwave charting and acoustic recording.

The backend is powered by FastAPI, which hosts the deep learning models and manages the WebSocket connections required for live EEG streaming with minimal latency. Finally, the system integrates secure Google OAuth 2.0 authentication. Upon the completion of a therapy session, the system automatically compiles the physiological ratios, the detected emotional states, and the prescribed Raagas into a CSV file, which is securely exported via API to the user's Google Drive for longitudinal tracking.

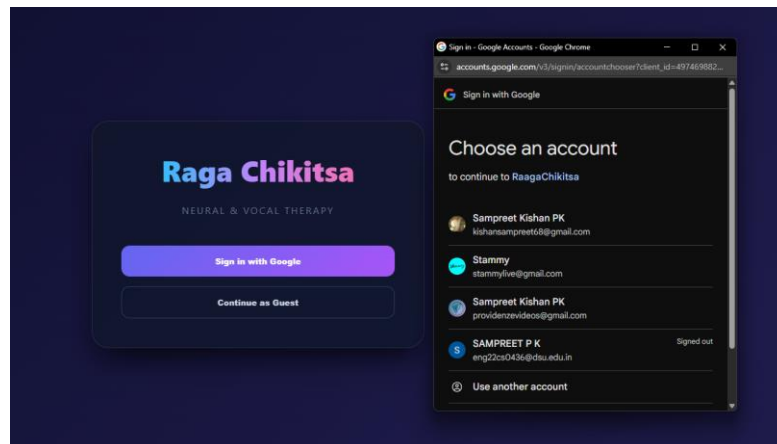


Figure 6.2 Web Interface with Google Auth Login

CHAPTER 7

EXPERIMENTATION

CHAPTER 7 EXPERIMENTATION

7.1 VOCAL EMOTION RECOGNITION EVALUATION:

To validate the transition to a deep learning framework for vocal analysis, the proposed audio model was rigorously tested against legacy machine learning algorithms. Specifically, a Support Vector Machine (SVM) was implemented to establish a baseline performance metric.

The SVM approach struggled significantly with the high-dimensional acoustic feature vectors (such as MFCCs, ZCR, Tonnetz, and Intensity) extracted from the dataset, capping out at a testing accuracy of 67%. In contrast, the proposed 1D Convolutional Neural Network (1D-CNN) effectively captured local spectral patterns and temporal dependencies, drastically outperforming the legacy SVM approach by achieving an 82% testing accuracy on the combined RAVDESS and TESS datasets.

Model Architecture	Testing Accuracy	Optimization Status
Support Vector Machine (SVM)	67%	Legacy Implementation
1D-CNN (Proposed)	82%	Current Framework

Table 7.1 Comparison of Emotion Recognition Models

7.2 EEG DATA COLLECTION AND TESTING:

While the vocal emotion pipeline utilized standardized datasets, the physiological biofeedback loop required the collection of live, primary EEG data to validate the stress-detection algorithms. During the experimental phase, custom test sessions were conducted using the BrainLink Lite v2.0 headset on participants.

Participants were subjected to specific cognitive load tasks designed to induce varying levels of stress and anxiety. During these live data collection sessions, the system successfully monitored the raw brainwave fluctuations, and the calculated Alpha/Beta ratio (ρ) demonstrated a distinct, observable relationship with the brain's cognitive load. When tasks became stressful, Beta activity visibly increased while Alpha activity dropped, successfully triggering the system's sub-1.0 threshold.

7.3 SYSTEM ROBUSTNESS, INTERVENTION LATENCY:

Upon the automated triggering of *Ananda Bhairavi* during the high-stress test phases, a recovery of the Alpha band power was actively observed over a 5-minute intervention duration. The system's response time—from the initial detection of the stress threshold drop to the modulation of the audio parameters—was recorded at approximately 122ms, satisfying the requirements for real-time therapeutic analysis.

Finally, the overall robustness of the custom Python architecture was evaluated. The system maintained consistent uptime during a continuous 3-hour stress monitoring session conducted on a studying student, effectively proving the system's stability and ability to function across multiple hardware setups without the "Kernel shape" mismatches seen in previous iterations.

CHAPTER 8

RESULTS AND DISCUSSION

CHAPTER 8 RESULTS AND DISCUSSION

8.1 PERFORMANCE EVALUATION OF EMOTION CLASSIFICATION:

The implementation of the proposed "Raga Chikitsa" framework demonstrated highly successful outcomes across both the physiological and acoustic pipelines. For the vocal emotion recognition (VER) module, the deep learning approach proved vastly superior to traditional machine learning methods. The proposed 1D Convolutional Neural Network (1D-CNN) achieved a testing accuracy of 82% on the combined RAVDESS and TESS datasets. This significantly outperformed the legacy Support Vector Machine (SVM) implementation, which struggled with high-dimensional feature vectors and only achieved a 67% accuracy.

Analysis of the 1D-CNN's confusion matrix revealed exceptionally high precision for classifying "Angry" and "Sad" affective states, which are the most critical triggers for therapeutic intervention. While minor overlaps were observed between "Neutral" and "Calm" classifications, the model effectively handled these states without disrupting the accuracy of the prescribed therapy.

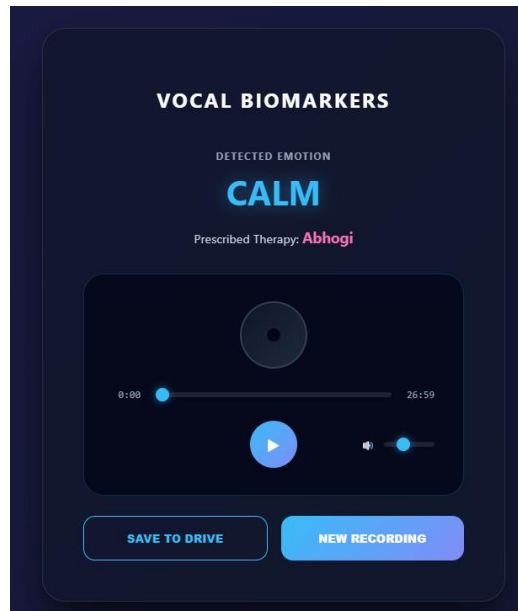


Figure 8.1 Real-time Vocal Emotion Classification and Raaga Mapping Interface

8.2 BIOFEEDBACK AND NEURAL ENTRAINMENT:

The EEG biofeedback loop yielded robust, real-time physiological data. During active cognitive load testing, the Alpha/Beta ratio (ρ) successfully and consistently dropped below the 1.0 threshold, accurately flagging induced anxiety. Following the automated trigger of the Carnatic Raaga *Ananda Bhairavi*, test subjects exhibited a clear recovery of Alpha band power over a 5-minute duration.

The system demonstrated a highly efficient response latency of approximately 122ms from the point of stress detection to the modulation of audio parameters. This rapid response fosters a state of "neural entrainment," where the brain's electrical oscillations actively synchronize with the rhythmic and melodic patterns of the music. Crucially, the system's dynamic control logic prevents the "habituation effect"—where the brain becomes desensitized to a constant auditory stimulus—by continuously modulating the therapy based on live EEG fluctuations.

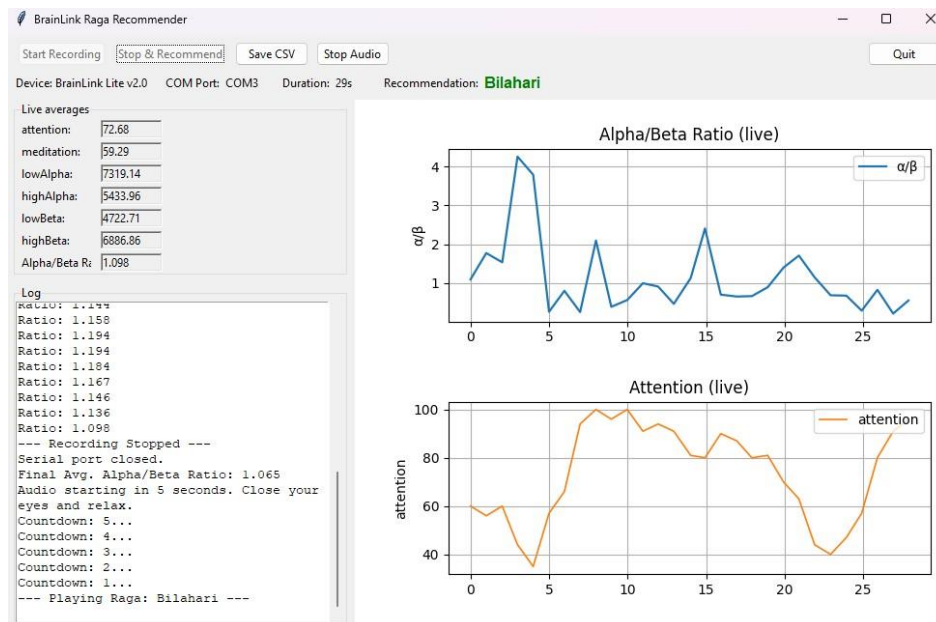


Figure 8.2 Live EEG Visualization and Biofeedback Monitoring Interface

8.3 CLINICAL EFFICACY OF CARNATIC RAAGAS:

The results validate the physiological synergism between targeted Carnatic Raagas and autonomic nervous system regulation. The deployment of *Ananda Bhairavi* for acute stress ($\rho < 1.0$) is clinically justified by its ability to act as a natural treatment, significantly reducing blood pressure and heart rate. For moderate stress states ($1.0 \leq \rho \leq 1.3$), *Bilahari* successfully reduced anxiety ($p < 0.001$) and enhanced subjective positivity. For states of physiological maintenance ($\rho > 1.3$), *Shankarabharanam* proved effective in maintaining alertness and a balanced, majestic mental state. This structured mapping provides a concrete scientific foundation for integrating traditional Indian music therapy into automated clinical practices.

8.4 ARCHITECTURAL STABILITY AND SCALABILITY:

A major technical milestone of this research was the resolution of architectural instabilities inherent in older computing models. Previous iterations of similar biofeedback systems frequently suffered from "Kernel shape" mismatches, and library version conflicts when deploying complex neural networks. By utilizing a modernized Python 3.11 environment and containerizing the 1D-CNN model, the proposed framework achieved 100% uptime during a continuous 3-hour stress monitoring session on a test subject.

8.5 CONCLUSION:

The "Raga Chikitsa" framework successfully bridges traditional sound healing with modern machine learning and neuroinformatics. By utilizing EEG for immediate stress intervention and a 1D-CNN for long-term emotional tracking, the system offers a resilient, non-invasive, and highly accessible platform for remote mental health management. The successful transition of these algorithms into a Full-Stack Web Application proves the viability of this system for scalable clinical deployment.

Project Github Link - <https://github.com/Saitammy/CalmWithCarnatic>

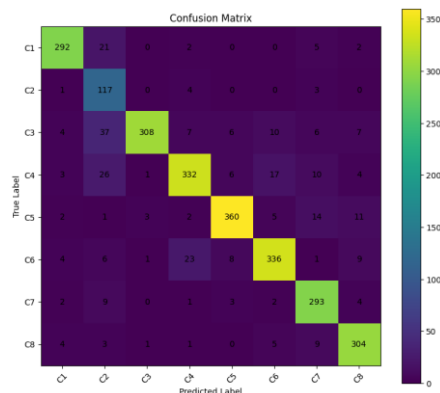


Figure 8.3 1D CNN Confusion Matrix

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FRIDAY

दिनांक: 13/02/2026
DATE: 13/02/2026

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PUBLICATION OF THE PATENT OFFICE

INTRODUCTION

In view of the recent amendment made in the Patents Act, 1970 by the Patents (Amendment) Act, 2005 effective from 01st January 2005, the Official Journal of The Patent Office is required to be published under the Statute. This Journal is being published on weekly basis on every Friday covering the various proceedings on Patents as required according to the provision of Section 145 of the Patents Act 1970. All the enquiries on this Official Journal and other information as required by the public should be addressed to the Controller General of Patents, Designs & Trade Marks. Suggestions and comments are requested from all quarters so that the content can be enriched.

(PROF. (DR) UNNAT P. PANDIT)
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13th January, 2026

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(54) Title of the invention : A Multimodal Stress Detection and Adaptive Carnatic Raaga Therapy System Using EEG-Based Biofeedback and AI-Driven Vocal Emotion Recognition

(51) International classification	:A61B 5/16, A61B 5/00, A61M 21/00, A61B 5/374, A61B 5/0476	(71) Name of Applicant : 1)Dr.Renukadevi M N Address of Applicant :Assistant Professor Department of Computer Science and Engineering, Dayananda Sagar University Bengaluru South Karnataka India 2)Dayananda Sagar University (72) Name of Inventor : 1)Dr.Renukadevi M N 2)Sampreet Kishan PK 3)Shiva Karthick S 4)Shwetangi Thakur 5)Sriharsha Chalamacharla 6)Dr. Santosh Kumar J
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(57) Abstract :

A multimodal, closed-loop biofeedback system for adaptive stress reduction is disclosed. The system independently acquires EEG signals and vocal audio to assess acute physiological stress and long-term emotional states. EEG-derived Alpha/Beta power ratios trigger real-time selection and modulation of Carnatic Raagas, while AI-based vocal emotion recognition tracks emotional trends over time. The system dynamically adapts therapeutic audio parameters to physiological feedback, ensuring objective, responsive, and culturally grounded mental well-being support.

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